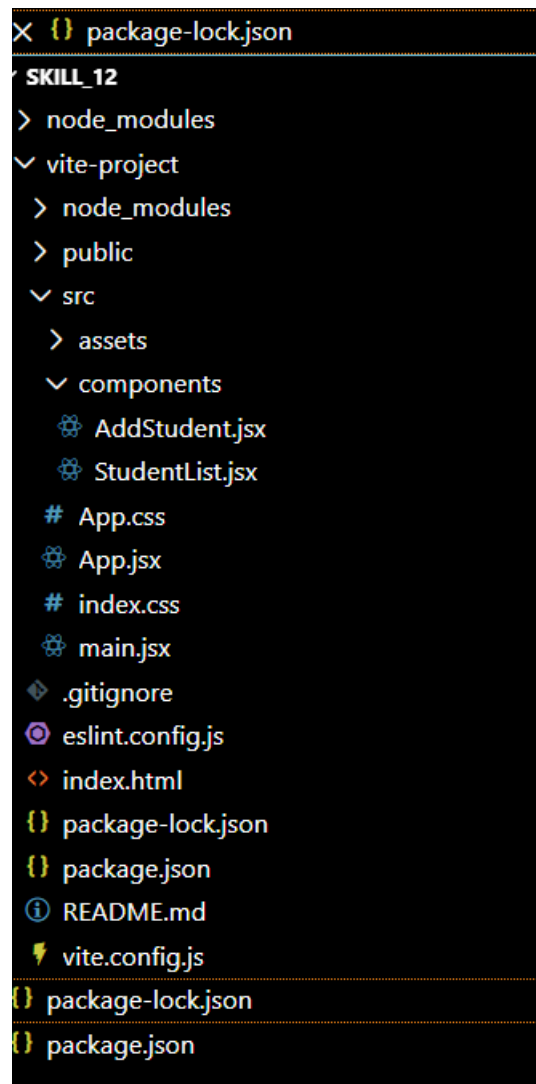


SKILL-12

Name: D.M Saifulla

ID:2400031125

Project structure:



StudentList.jsx

```

1  import React, { useEffect, useState } from 'react';
2  import axios from 'axios';
3
4  const StudentList = ({ onEdit }) => {
5      const [students, setStudents] = useState([]);
6
7      // Function to fetch students
8      const fetchStudents = async () => {
9          try {
10             const response = await axios.get("http://localhost:8080/students");
11             setStudents(response.data);
12         } catch (error) {
13             console.error("Error fetching students:", error);
14         }
15     };
16
17     useEffect(() => {
18         fetchStudents();
19     }, []);
20
21     const deleteStudent = async (id) => {
22         await axios.delete(`http://localhost:8080/students/${id}`);

```

AddStudent.jsx

```

1  import React, { useState, useEffect } from 'react';
2  import axios from 'axios';
3
4  const AddStudent = ({ selectedStudent, refreshList, clearSelection }) => {
5      const [student, setStudent] = useState({ name: '', email: '', course: '' });
6
7      // Fill form if editing
8      useEffect((parameter) selectedStudent: any
9      |   if (selectedStudent) setStudent(selectedStudent);
10     }, [selectedStudent]);
11
12     const handleChange = (e) => {
13         setStudent({ ...student, [e.target.name]: e.target.value });
14     };
15
16     const handleSubmit = async (e) => {
17         e.preventDefault();
18         if (student.id) {
19             // Update
20             await axios.put(`http://localhost:8080/students/${student.id}`, student);
21         } else {
22             // Create
23             await axios.post("http://localhost:8080/students", student);

```

App.jsx

```

import React, { useState } from 'react';
import StudentList from './components/StudentList';
import AddStudent from './components/AddStudent';

function App() {
  // Stores the student currently being edited
  const [selectedStudent, setSelectedStudent] = useState(null);

  // A simple toggle to force the StudentList to refresh data
  const [refreshKey, setRefreshKey] = useState(0);

  const handleRefresh = () => {
    setRefreshKey(prev => prev + 1);
  };

  const handleEdit = (student) => {
    setSelectedStudent(student);
    // Scroll to top so user sees the form is filled
    window.scrollTo(0, 0);
  };

  return (
    <div style={{

```

Package.json

```

{
  "name": "vite-project",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "axios": "^1.6.7",
    "react": "^18.2.0",
    "react-dom": "^18.2.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.9.0",
    "@types/react": "^18.3.3",
    "@types/react-dom": "^18.3.0",
    "@vitejs/plugin-react": "^4.3.1",
    "eslint": "^9.9.0",

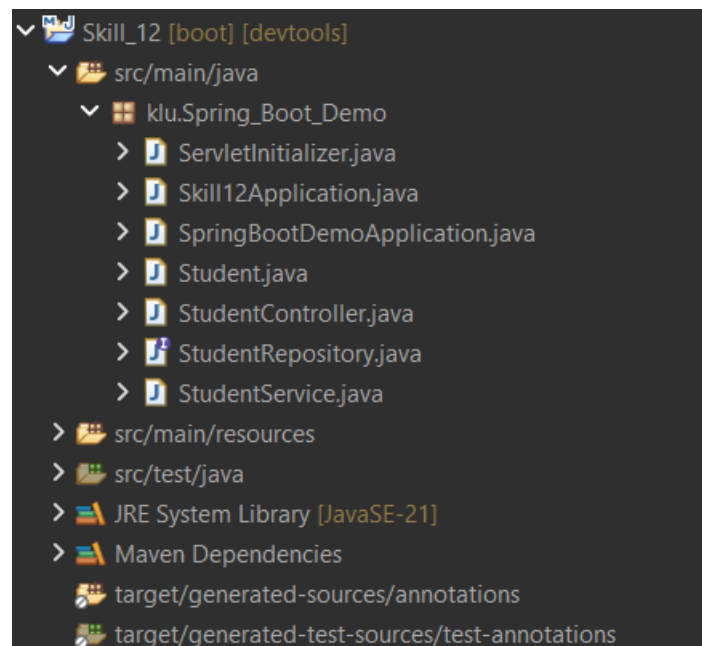
```

Packagelock.json:

```
{
  "name": "Skill_12",
  "lockfileVersion": 3,
  "requires": true,
  "packages": {
    "": {
      "dependencies": {
        "axios": "^1.13.6"
      }
    },
    "node_modules/asynckit": {
      "version": "0.4.0",
      "resolved": "https://registry.npmjs.org/asynckit/-/asynckit-0.4.0.tgz",
      "integrity": "sha512-Oei90H4tRh0YqU3GxhX79dM/mwVgVbZJaSNaRk+bshkj0S5cfHcgYakreBjrHwatXKb",
      "license": "MIT"
    },
    "node_modules/axios": {
      "version": "1.13.6",
      "resolved": "https://registry.npmjs.org/axios/-/axios-1.13.6.tgz",
      "integrity": "sha512-ChTCHMouEe2kn713WHbQGcuYrr6fXTBiu4600TWwWob16g1bXn4vtz070pe7ewMozJ",
      "license": "MIT",
      "dependencies": {
        "follow-redirects": "^1.15.11",

```

Project structure:



Student.java

```

package k10.Spring_Boot_Demo;

import jakarta.persistence.*;

@Entity
@Table(name = "student_table")
public class Student {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    private String name;
    private String email;
    private String course;

    // Getters and Setters
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }
    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }
}

```

StudentController.java

```

package k10.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@RestController
@RequestMapping("/students")
@CrossOrigin(origins = "http://localhost:5173") // Connects to your React app
public class StudentController {

    @Autowired
    private StudentService service;

    @GetMapping
    public List<Student> getAll() {
        return service.getAllStudents();
    }

    @PostMapping
    public ResponseEntity<Student> add(@RequestBody Student student) {
        return ResponseEntity.ok(service.saveStudent(student));
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(@PathVariable Long id, @RequestBody Student studentDetails) {
        Student student = service.getStudentById(id);
        if (student != null) {
            student.setName(studentDetails.getName());
            student.setEmail(studentDetails.getEmail());
            student.setCourse(studentDetails.getCourse());
            return ResponseEntity.ok(service.saveStudent(student));
        }
    }
}

```

StudentRepository.java


```

package klu.Spring_Boot_Demo;

import org.springframework.data.jpa.repository.JpaRepository;

@Repository
public interface StudentRepository extends JpaRepository<Student, Long> {
}

```

StudentService.java

```

package klu.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@Service
public class StudentService {
    @Autowired
    private StudentRepository repository;

    public List<Student> getAllStudents() {
        return repository.findAll();
    }

    public Student saveStudent(Student student) {
        return repository.save(student);
    }

    public void deleteStudent(Long id) {
        repository.deleteById(id);
    }

    public Student getStudentById(Long id) {
        return repository.findById(id).orElse(null);
    }
}

```

SpringBootDemoApplication.java

```

package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class SpringBootDemoApplication {
    public static void main(String[] args) {
        // This line tells Spring Boot to start the whole application
        SpringApplication.run(SpringBootDemoApplication.class, args);
    }
}

```

Skill12Application.java

```

package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class Skill12Application {

    public static void main(String[] args) {
        SpringApplication.run(Skill12Application.class, args);
    }

}

```

The screenshot displays an IDE environment with the following components:

- Project Explorer (Left):** Shows a project named 'demo_project' with a package 'klu.Spring_Boot_Demo'. The package contains several Java files, including 'Skill12Application.java'.
- Editor (Center):** Displays the source code of 'Skill12Application.java', which is identical to the code shown in the previous blocks.
- Console (Bottom):** Shows the output of the application. The logs indicate that the application started successfully, with messages from Spring Boot and HikariPool.

The console output includes the following lines:

```

2026-03-17T12:22:52.292+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext
2026-03-17T12:22:53.098+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] org.hibernate.orm.jpa : HHH000540: Processing Pers
2026-03-17T12:22:53.456+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] org.hibernate.orm.core : HHH000001: Hibernate ORM d
2026-03-17T12:22:55.215+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup:
2026-03-17T12:22:55.315+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-03-17T12:22:55.882+05:30 INFO 11816 --- [Spring_Boot_Demo] [ restartedMain] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added conn

```

```
1  {
2    "name": "vite-project",
3    "private": true,
4    "version": "0.0.0",
5    "type": "module",
6    "scripts": {
7      "dev": "vite",
8      "build": "vite build",
9      "lint": "eslint .",
10     "preview": "vite preview"
11   },
12   "dependencies": {
13     "axios": "^1.6.7",
14     "react": "^18.2.0",
15     "react-dom": "^18.2.0"
16   },
17   "devDependencies": {
18     "@eslint/js": "^9.9.0",
19     "@types/react": "^18.3.3",
20     "@types/react-dom": "^18.3.0",
21     "@vitejs/plugin-react": "^4.3.1",
22     "eslint": "^9.9.0",

```

esbuild - vite-project + v

PS C:\Users\LAASYA\Desktop\Skill_12\vite-project> npx vite

VITE v7.3.1 ready in 860 ms

- Local: <http://localhost:5173/>
- Network: use --host to expose
- press h + enter to show help

OUTPUT:

Student Management System

Add Student

			<input type="submit" value="Submit"/>
----------------------	----------------------	----------------------	---------------------------------------

Student List

Name	Email	Course	Actions
------	-------	--------	---------